

Abhiney Sharma

DNS ADMINISTRATION

COMPREHENSIVE STUDY NOTES
(Solution Architect Level)



HIGHLIGHT: FROM ZERO TO HERO:

Basics, Hierarchy, Zones,
Security

✓ Advanced Troubleshooting



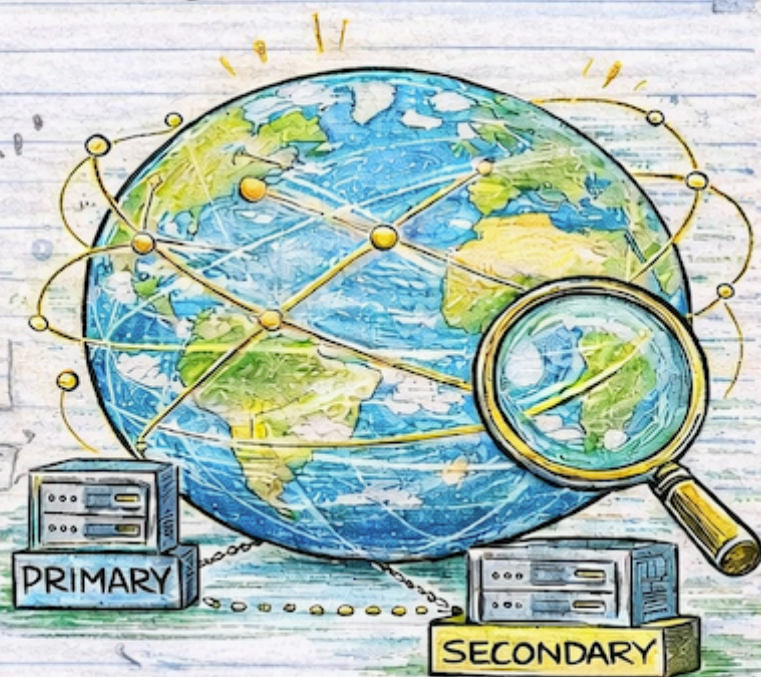
OWNER DETAILS

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"The Phonebook of the Internet."



The Foundation - What are we managing?

1. What is DNS? (The Admin Perspective)

- User's View: "Magic that turns **google.com** into a website."
- Admin's View: A **distributed, hierarchical database** that maps specific keys (names) to values (IPs, text, services).
- Protocol: Layer 7 (Application) 
- Ports:
 - **UDP 53**: Standard queries (fast, low overhead).
 - **TCP 53**: Zone transfers (replication) or large responses (>512 bytes).



2. Why do we need Admins?

DNS is not "set and forget."

① **Availability**: If DNS dies, the internet "breaks" for your company.



② **Performance**: Latency in DNS = slow website loading.



③ **Accuracy**: Wrong IP = Downtime.



It's always DNS.

In network troubleshooting, 90% of "connectivity" issues are actually "name resolution" issues.

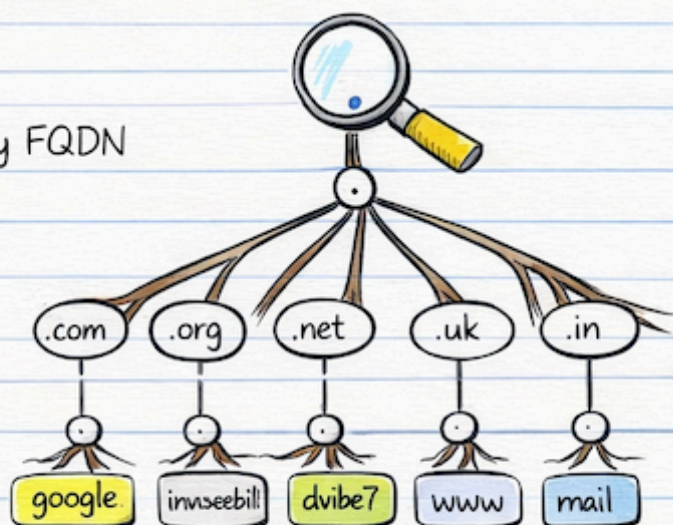


The Hierarchy (The Map)

The DNS namespace is an Inverted Tree.

1. The Root (.)

- The silent anchor at the end of every FQDN (Fully Qualified Domain Name.).
- **www.example.com.** ←← That dot!
- Managed by **ICANN/IANA**.
- **(13)** Logical Root Clusters (A-M).



2. TLD (Top-Level Domain)

- **gTLD (Generic):** .com, .org, .net
- **ccTLD (Country):** .in, .uk, .jp
- **Management:** These are managed by "Registries" (e.g., Verisign manages .com).



3. Second-Level Domain (SLD)

- **google.com**, **invoicebill.org**, **dvibe7.com**
- **THIS IS YOUR DOMAIN.**
- You rent this space. You are the "Dictator" of this zone.



MEMORY TRICK:

Read domains Right-to-Left to find the boss.

- (Root) → org (Registry) → invoicebill → (Abhiney) → www (Server)



Read domains **Right-to-Left** to find the boss.

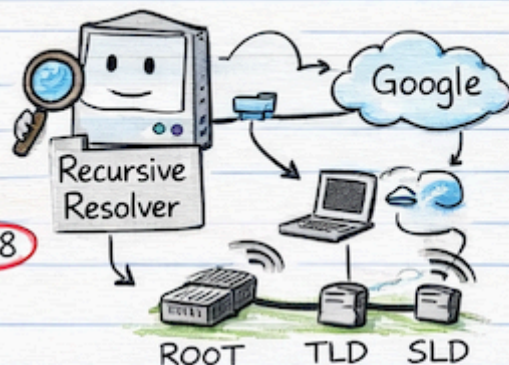


Server Roles - The Hardware

Not all DNS servers do the same job. Distinct roles prevent confusion.

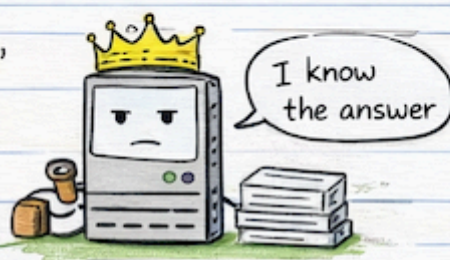
1. The Recursive Resolver (The Client's Friend)

- Role: "I don't know the answer, but I will go find it for you."
- Client: Your laptop/phone talks to "this" server (usually provided by ISP or Google **8.8.8.8**)
- Behavior: Caches answers aggressively to speed up browsing.



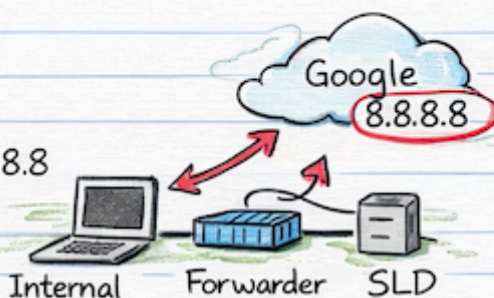
2. The Authoritative Server (The Boss)

- Role: "I know the answer because I own the data."
- Client: Does NOT talk to end-users directly usually; talks to Resolvers.
- Behavior: Does not cache. It holds the "Master File."



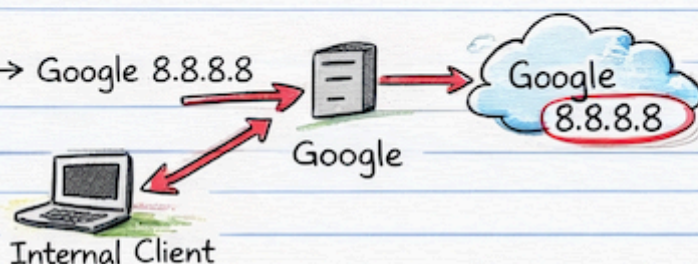
3. The Forwarder

- A middle-man.
- Internal Client → Forwarder → Google 8.8.8.8
- Used to reduce traffic on the WAN link.



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- A middle-man.
- Internal Client → Forwarder → Google 8.8.8.8



How a Query Works - The Flow

Scenario: Client requests www.example.com.

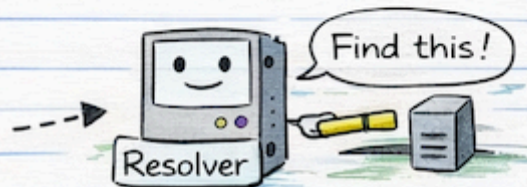
1. Local Check:

- Browser Cache → OS Cache (hosts file)



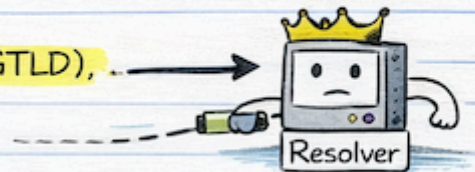
2. Recursive Query:

- Client asks Resolver: "Find this!"



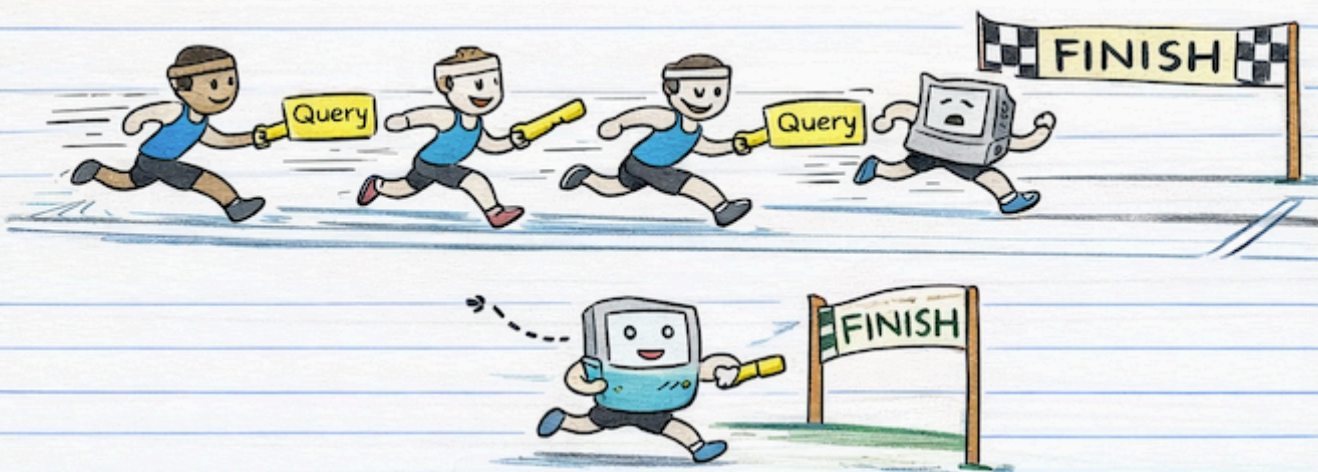
3. Iterative Process (The Resolver's Hunt)

- Resolver → Root Server: "Where is .com?"
- Root → TLD Server: "Go ask the .com server (GTLD),"
- TLD → TLD Server: "Where is example.com."
- Resolver → Auth Server: "Go ask ns1.example.com."
- Auth Server → "Yes. IP is 1.2.3.4."
- Resolver → Do you have www?"



4. Final Step:

- Resolver caches it and sends to Client.



The Zone File – The Database

As an Admin, you manage Zones. A Zone is a text file containing records.

1. Forward Lookup Zone

- Purpose: Maps → IPs
- Most common zone type.
- Example: "What is the IP of mail.google.com?"



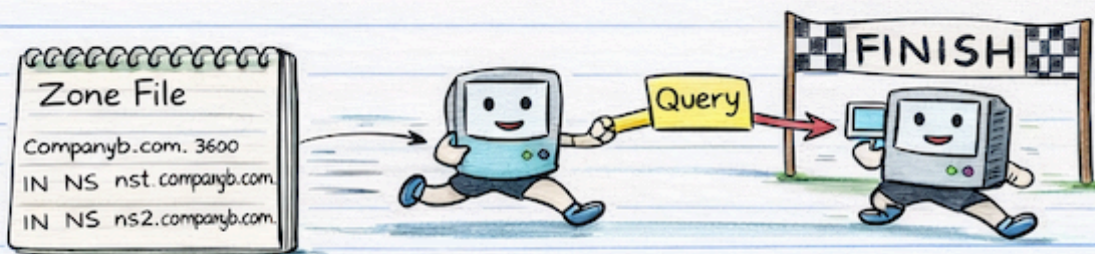
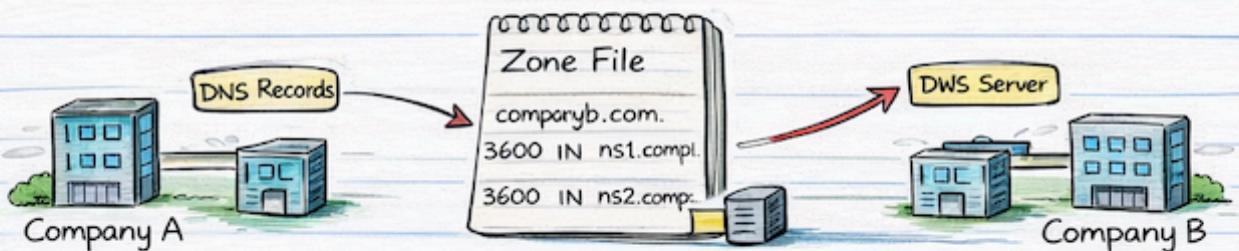
2. Reverse Lookup Zone

- Purpose: Maps IPs → Names
- Critical for security logs and email servers.
- * Naming: x.x.x.in-addr.arpa



3. Stub Zone

- A "Shortcut" zone.
- It contains ONLY the Nameserver (NS) records of another domain.
- * Use Case: Fast forwarding between two company branches.



Key Record Types – Part I

The standard bricks of the DNS wall.

Record	Name	Purpose	Example
A	Address	IPv4 Mapping	host → 192.168.1.1 
AAAA	Quad-A	IPv6 Mapping	host → 2001:db8::1 
CNAME	Canonical Name	Alias (Nickname)	www → host.example.com 

1. Forward Lookup Zone

- Purpose: Maps IPs → Names:
- Critical for security logs and email servers.
- * Naming: x.x.x.in-addr.arpa



3. CNAME

- Alias (Nickname)



! WARNING:

CNAME Restrictions:

1. You cannot have a CNAME at the root (@).
2. If a name is a CNAME, it cannot have "any" other record types (no MX, no TXT).

Key Record Types – Part II

The "Service" records.

1. MX (Mail Exchange)

- Tells the internet where to send email for your domain.
- **Priority**: Lower number = Higher Priority.
 - **10** mail.google.com (Primary)
 - **20** backup.google.com (Secondary)



10 mail.google.com

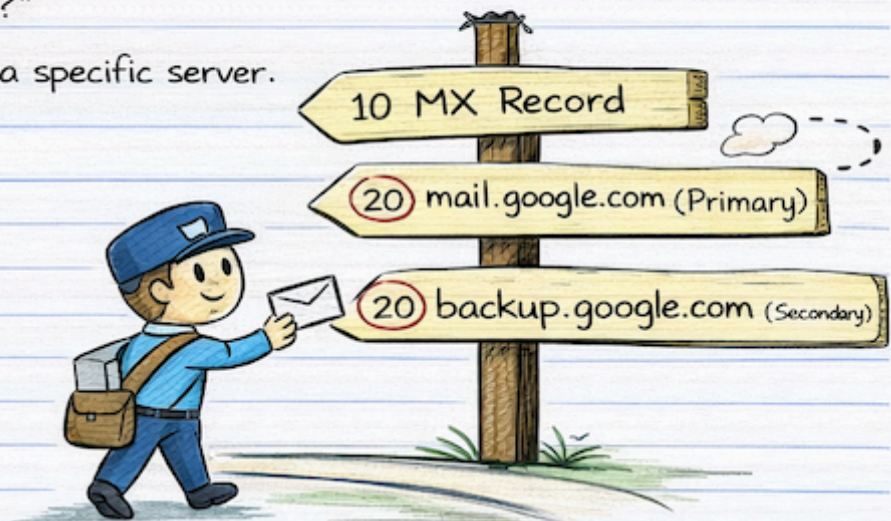
2. TXT (Text)

- Originally for notes, now used for **Verification**.
- **SPF** (Sender Policy Framework): "Only IP X is allowed to send email for me."
- **Verification**: "Prove you own this domain for Google Console."



3. NS (Nameserver)

- "Who is in charge here?"
- Delegates authority to a specific server.

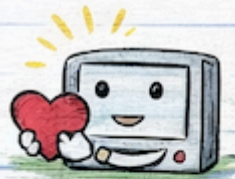


The SOA Record – The Heart

Every Zone **MUST** start with a **Start of Authority (SOA)** record.

It sets the rules for the zone.

Plaintext



```
@ IN SOA ns1.example.com. admin.example.com. (  
    2026011601 ; Serial Number  
    3600       ; Refresh  
    1800       ; Retry  
    604800     ; Expire  
    86400      ) ; Minimum TTL
```



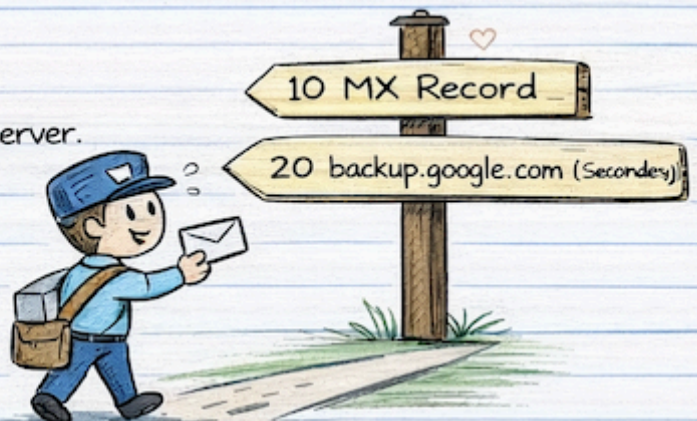
- **MNAME**: Primary Name Server.
- **RNAME**: Email of Admin (Note: the @ is replaced by a ".").
- **Serial**: **CRITICAL!**

If you change data but forget to increase this number, Secondary servers will NEVER update.



3. NS (Nameserver)

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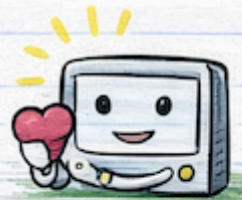


Recommended Serial Format: **YYMMDDnn**
(Year, Month, Day, Revision Number.)

Primary vs. Secondary Architecture

Redundancy is King.

You never run just one DNS server.



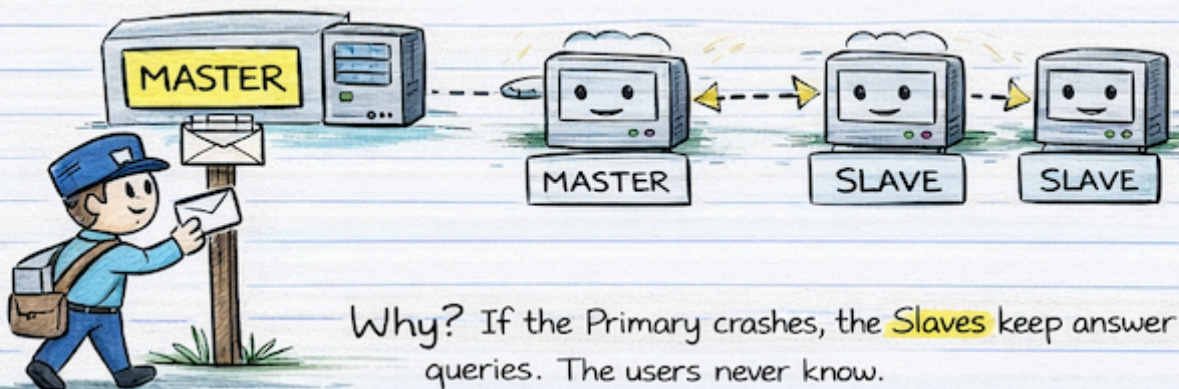
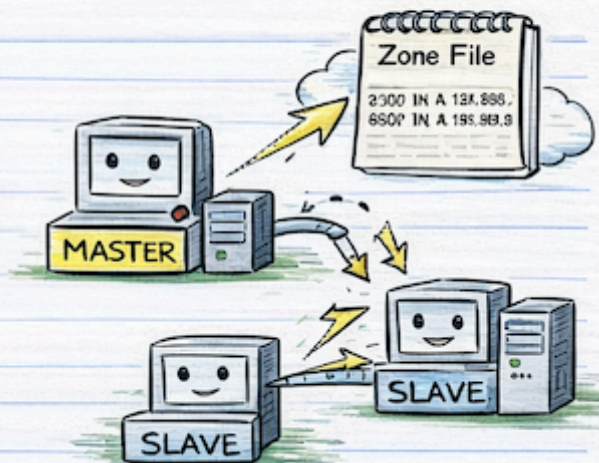
1. Primary (Master)

```
@ IN SOA ns1 example.com. admin.example.com. (  
  2026011601 ; Serial Number  
    3600      ; Refresh  
    1800      ; Retry  
   604800     ; Expire  
   86400      ) ; Minimum TTL
```



2. Secondary (Slave)

- The Read-Only copy.
- You cannot edit files here.
- It downloads data from the Primary automatically.
- You can have unlimited Secondaries.



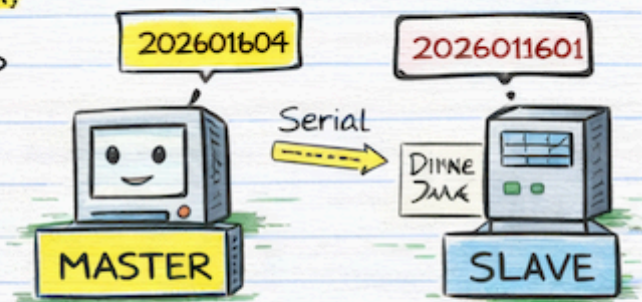
Why? If the Primary crashes, the Slaves keep answering queries. The users never know.

Zone Transfers (Replication)

How the Slave learns, from the Master.

The Process

- 1 Slave checks Master's **SOA Serial**,
- 2 Is Master Serial > Slave Serial?
- 3 YES: Request Update.
- 4 NO: Do nothing.



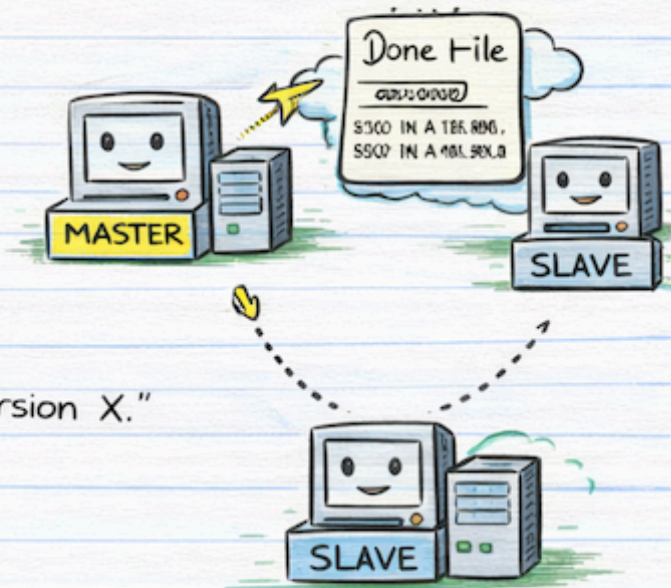
Types of Transfer:

- **AXFR (Full Transfer)**

"Give me the WHOLE database."
(High Bandwidth)

- **IXFR (Incremental Transfer)**

"Just give us the changes since version X."
(Efficient)



SECURITY CHECK:

Hackers love AXFR. If they can download your zone, they know every server IP in your company.

Config: `allow-transfer { none; };` (Only allow specific **Slave IPs**).

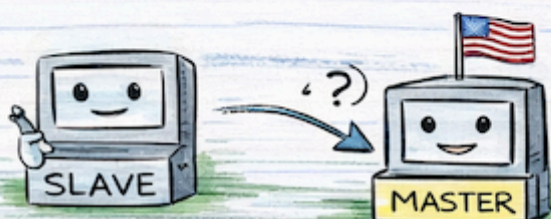
Delegation & Glue Records

Delegation

Breaking a large domain into smaller pieces managed by different teams.

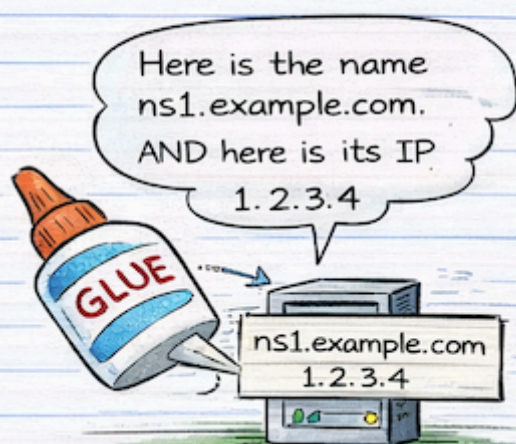
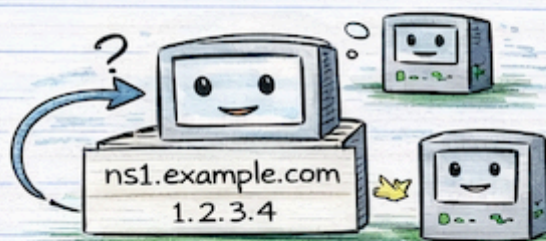
The Process

1. Slave checks Master's SOA Serial.
2. usa.company.com (HQ IT Team)



The Glue Record Problem

- If ns1.example.com is inside example.com... how do you find ns1?
- Circular Dependency?
- Solution: A Glue Record.
- You tell the TLD (Registry):
"Here is the name ns1.example.com;
AND here is its IP 1.2.3.4."



SECURITY CHECK:

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Split-Horizon DNS – Enterprise Level

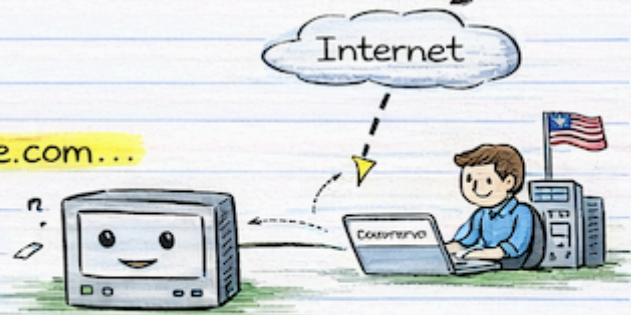
Serving different answers to different people for the **same name**.

Scenario

- Resource: **crm.company.com**
- **Internal User**: Should go to LAN IP **10.5.5.5** (Fast, Direct)
- **External User**: Should go to Public IP **203.0.113.5** (VPN/Firewall)

The Glue Record Problem

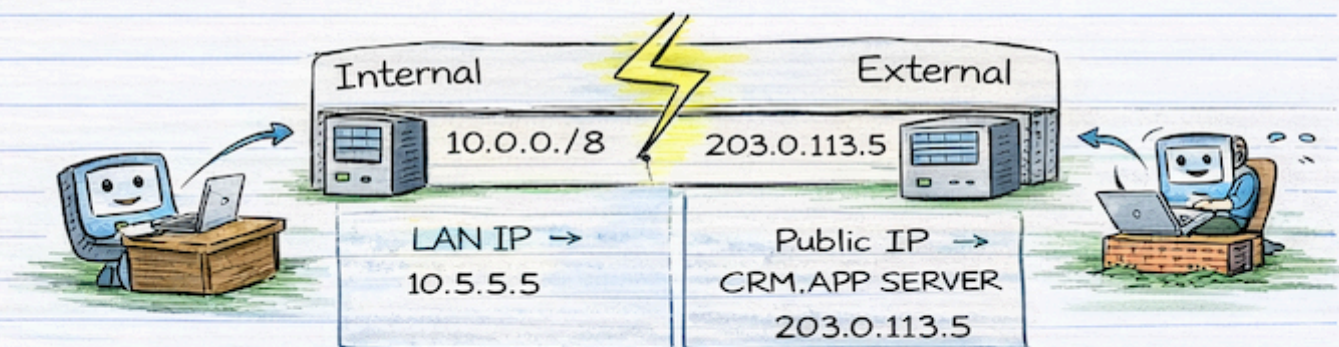
- If **ns1.example.com** is inside **example.com**...
how do you find **ns1**?



Implementation (Views)

Using "Views" in BIND config:

- View "Internal": Matches clients **10.0.0.0/8**
- View "External": Matches **any**



This allows you to hide sensitive **internal server names/IPs** from the public internet!

13. DEFENSIVE ARCHITECTURE

The tools you need to survive.

1. RESTRICTING RESOLVERS

- **Issue:** An "Open Resolver" answers queries for the entire internet
- **Attack:** Causes high traffic volume and resource exhaustion (Traffic Amplification)
- **Fix:** Configure Access Control Lists (ACLs).

```
allow-query { localnets; };
```



2. DATA INTEGRITY (Anti-Spoofing)

- **Issue:** "Cache Pollution" (False data injected into the server)
- **Fix:** DNSSEC (Security Extensions).
- **Logic:** Adds a "Digital Signature" to records.
 - **Analogy:** Like a wax seal on a letter.
It proves the data is authentic and untouched.



3. HIDDEN MASTER TOPOLOGY

- **Strategy:** The Public only sees the **Secondary (Slave)** servers.
- **Benefit:** The Primary (Master) is isolated behind a firewall.
 - Protects the "Source of Truth" from direct access.

3. HIDDEN MASTER TOPOLOGY

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14. TROUBLESHOOTING TOOLKIT

The tools you need to survive.

1. nslookup

- Old, Windows-based, interactive.
- Good for quick checks. `> server 8.8.8.8` to switch targets.



2. dig (Domain Information Groper)

- The Pro Tool. Linux/Mac.



- Flags:
 - +short: Just the IP
 - +trace: Show the full iterative path from Root.
- @server: Query a specific server.
- AXFR: Attempt a zone transfer.

3. host

- Simple. `host google.com`.



Final Mnemonic for Troubleshooting:

"Does It Go?" (DIG) →

→ Always use DIG to see the raw data!

15. THE ARCHITECT'S CHEATSHEET

The tools you need to survive.

A. THE VITAL NUMBERS

- **Port 53 (UDP):** Standard Queries (Small, Fast)
- **Port 53 (TCP):** Zone Transfers & Large Responses (>512 Bytes)
- **TTL (Time To Live):** How long to cache.
 - High (24h) = Static Records.
 - Low (60s) = Failover/Changing Records.



B. RECORD TYPES (The Big 5)

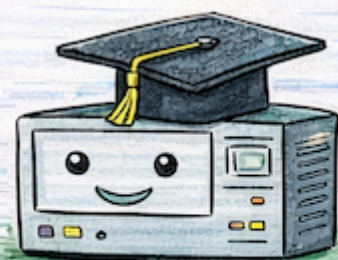
1. **A / AAAA:** Name → IP (v4/v6).
2. **CNAME:** Alias → Canonical Name. (Never at Root!)
3. **MX:** Mail → Server. (Lower # = Higher Priority)
4. **NS:** Zone → Authoritative Server.
5. **TXT:** Text → Verification/SPF.

C. COMMAND LINE SURVIVAL

- **dig +short google.com** → Just the IP.
- **dig +trace google.com** → See the full hierarchy walk.
- **dig @server name** → Query a specific DNS server.
- **rndc reload** → Apply config changes without restart.

D. THE GOLDEN RULES

1. **Redundancy:** Always have at least "2" Name Servers.
2. **Serial #:** Always INCREMENT the SOA Serial after editing.
3. **Security:** Block AXFR (Zone Transfer) to the public.
4. **Glue:** Required when the NS is "inside" the zone it manages.




FINAL THOUGHT:

END OF NOTES